

Dilawar Singh

Senior Systems Engineer | Kernel, Security & Platform Engineering

+91-9108-750-527 | dilawar.s.rajput@gmail.com | dilawar.in | github.com/dilawar

Founding / Senior Systems Engineer with 10+ years of experience building kernel-level components (eBPF, Windows minifilters), endpoint and data security systems, SaaS platforms, and real-time simulators. Proven record of taking deep-tech products from PoC to production: designing core architectures, writing performance-critical low-level code in Rust and C/C++, and shipping cross-platform systems. Enthusiastic contributor to Free and Open Source Software (FOSS).

Core Competencies: Kernel Development | eBPF | Windows WDK | Rust | C/C++ | Endpoint Security | Real-time Systems | Systems Architecture | CI/CD | Team Leadership

Skills

- Systems & Devices: SDK development (Rust, C++), Linux Kernel Modules, Extended Berkeley Packet Filter (eBPF), Windows Kernel Minifilter (Windows Driver Kit/WDK), Win32 and POSIX APIs, IPC, GUI (Qt, ImGui), CLI tooling (Python, Rust, Shell)
- Languages: Rust, C, C++, Python (expert); Java, TypeScript, JavaScript, Nim, Haskell (intermediate); VHDL, Verilog, SystemC, BlueSpec (familiar); PHP
- SaaS Platforms & APIs: High-performance gRPC and REST services using FastAPI, Axum, Actix; SQL, PostgreSQL, DuckDB; job queues; OpenTelemetry, Grafana; data parsing, validation, and lifecycle management
- Infrastructure & DevOps: Docker, QEMU; CI/CD (GitLab, Jenkins); build systems; packaging (OBS, NSIS); Ansible, Pulumi; custom tooling and CI/CD pipeline design
- Scientific & Computation: Boost, GNU Scientific Library (GSL), SciPy stack, OpenCV; numerical simulators; computation library development; pattern matching and optimization (network flows, graphs, linear algebra); stochastic DSA; parallelism partitioning
- Other: Parser development, event-driven systems, technical literature review

Work Experience

Independent Consultant — Self-Employed, Bengaluru

Mar 2026 – Present

- Designing and developing scalable microservices supporting a virtual assistant platform for an automotive control system (Rust, C, Qualcomm SoC)
- Designing and overseeing development of high-performance APIs (Rust, AMQP) for a cyber lab running near-real-time experiments on highly correlated, diverse datasets (audio, video, sensor events)

Senior Engineer — Dagnosis, Bengaluru

Jan 2025 – Feb 2026

- Aggregated ~30 GB/session and ~10,000 events/session of live sensor data into a single operational control plane used for real-time decision-making and incident response
- Set up and operated self-hosted infrastructure (GitLab, PostgreSQL, Grafana, CI/CD runners), including access control, backups, and deployment policies

- Established risk-focused planning and sprint rituals that reduced invisible work, clarified ownership, and gave leadership visibility into team throughput
- Built hiring pipelines and hired 5 engineers, scaling the team from 6 to 11 in 3 months

Cofounder & CTO — Subconscious Compute, Bengaluru

Dec 2019 – Dec 2024

- Wrote Windows kernel minifilter (WDK, C++/Rust) and Linux eBPF sensors generating real-time telemetry for Shepherd's cross-platform threat detection and compliance engine, deployed across customer endpoints
- Hired and led a team of 15 systems engineers to build Shepherd, an automated endpoint compliance and security platform
- Authored technical memos and pitch decks to secure accelerator placements and support fundraising

Research Fellow & Google Summer of Code (GSoC) Mentor

2016 – 2018

- Added high-performance numerical solvers (C++, Boost, GNU Scientific Library) to the MOOSE simulator, enabling it to solve multi-scale neural simulation problems
- Wrote Python bindings (pybind11) and created packages for MOOSE and Smoldyn, significantly improving usability and adoption by the scientific community
- Mentored multiple Google Summer of Code (GSoC) contributors on GPU/CUDA-based acceleration over 4 years

Firmware Engineer — Kritikal Solutions, Noida

Jul 2009 – Jun 2010

- Implemented a Kalman-filter-based image stabilization system for professional movie cameras (C++, multi-threaded RTOS) to eliminate vibration artifacts from recordings

Projects

- Published Rust crates for stream processing, online statistics, Windows Management Instrumentation (WMI) interfaces, and CLI tooling
- Developed Python tools for scientific data extraction, instrumentation UIs, computer vision, and internal developer tooling

Education

- PhD, Computational Neuroscience — NCBS Bangalore (TIFR Mumbai), 2019
- Doctoral Studies, Digital Systems — IIT Bombay, 2011–2013
- M.Tech, Electrical Engineering (VLSI) — IIT Bombay, 2009
- B.Tech, Instrumentation & Control Engineering — Dr. MGR ERI Chennai, 2007

Publications

[Full list on Google Scholar](#)

- [Python interfaces for the Smoldyn simulator](#), Bioinformatics, 2022
- [BioSimulators: a central registry of simulation engines](#), Nucleic Acids Research (NAR), 2022